Project Scenario

The organising committee of the 2026 FIFA World Cup requires a database system to manage information about the tournament. The tournament will be hosted across United States, Mexico, and Canada. The organisers require a centralised SQL Server database that stores and manages:

- Participating national teams
- Players and coaching staff
- Stadiums and host cities
- Matches played from the group stage through to the final
- Ticket bookings for fans attending matches

The database must allow tournament administrators to track matches, teams, player statistics, ticket allocations, and fan attendance. Efficient data retrieval, integrity, and security are essential for ensuring reliable tournament operations. Students must design and implement a SQL Server database that stores and manages tournament information for the 2026 FIFA World Cup hosted in USA, Canada, and Mexico.

Your task is to design and implement a relational database using Microsoft SQL Server and T-SQL that supports these tournament management requirements. This project focuses strictly on database design and T-SQL implementation. No user interface, web application, or front-end development is required.

Project Overview

Students must design and implement a SQL Server relational database that manages tournament operations including:

- Teams and players
- Stadiums and host countries
- Matches from group stage to final
- Ticket bookings and fan attendance

The project consists of two main components:

1) **Theory Document (Database Design)**
2) **Practical Implementation (T-SQL Programming)**

The final submission will occur at the end of week 4.

Deliverables

Submit the following three files in one compressed or zip folder. Students must submit the following:

Part 1 – Theory Component (Database Design)
Students must submit a formal document explaining the database design process:

a) **Business Rules**: Define the business rules governing the tournament database.

Students must identify and document at least 10–15 business rules based on the scenario.

b) **Database Normalization**: Students must demonstrate the normalization process by showing how the database progresses through Unnormalized Form (UNF) to Third Normal Form (3NF). The final schema must clearly demonstrate that all tables are in 3NF. Students must focus on:

- Removal of repeating groups
- Elimination of partial dependencies
- Removal of transitive dependencies

c) **Entity Relationship Diagram (ERD)**: Students must design a complete ERD representing the tournament database.

Part 2 – Practical Implementation (T-SQL Programming)

Students must implement the database using Transact-SQL (T-SQL) in Microsoft SQL Server. All code must be provided in one structured SQL script file. The script should be organized into logical batches so the entire database can be created by executing the script once.

Required T-SQL concepts include the following:

- **Database Definition (using DDL)**
- **Database Manipulation (using DML)**
- **Database Control (DCL)**
- **Database Backup**

Mark Allocation

Criteria	Marks
Theory (PDF Document)	
Business Rules & Scenario Understanding	5
Normalization to 3NF	5
ERD Design Accuracy	5
ERD Relationships	5
T-SQL Script	
Database Creation	5
Table Population	5
Queries:	
Joins	5
Subqueries	5
CTEs	5
CASE	5
Programmability:	
Procedures	5
Functions	5
Views	5
Advanced T-SQL Concepts:	
Triggers	5
Cursors	5
Transactions	5
Security:	
Authentication	5
Encryption	5
Code Quality, Comments & Naming Conventions	5
Database Backup File (.bak)	
Backup	5

Rating Scale Definition

Score	Performance Level	Description
5	Excellent	Demonstrates outstanding understanding, correct implementation, and professional-quality work with no significant errors.
4	Very Good	Strong understanding with minor errors. Most requirements implemented correctly.
3	Satisfactory	Adequate understanding. Core requirements implemented but several issues exist.
2	Limited	Partial implementation with major conceptual or technical weaknesses.
1	Poor	Minimal evidence of understanding. Work largely incorrect or incomplete.
0	No Attempt/Submission	Criterion not attempted or not submitted.

Additional Information

- This is a group assignment. Groups may not exceed 4 people. Groups will be auto assigned.
- Names of all group members must appear on the cover page.
- Submit your assignment in one compressed or zip folder on BC Connect before the due date.
- All information, diagrams and SQL code must be your own original work.
- Plagiarism is a serious offence. Belgium Campus uses software that can scan for plagiarism and a student caught doing this will get 0 for this assignment.
- No mark will be awarded if the assignment is not uploaded via BC Connect.
- Late assignments will not be accepted; missing the deadline is an automatic 0.
- Any form of cheating will be taken seriously and may result in disciplinary action including disqualification from future assessments.
- **Auto-generated scripts or using Design View will not be acceptable under any circumstances**.